

1. SELECT DISTINCT (department)
FROM students ;

department
IT
HR
Finance

2. SELECT department, AVG (age) AS AVG-age
FROM students
GROUP BY department ;

department	AVG-age
IT	20,5
HR	22.
Finance	23

3. SELECT department, COUNT (student-id) AS student-count
FROM students
GROUP BY department ;

department	student-count
IT	2
HR	2
Finance	1

* HAVING student-count > 1 ;

department	student-count
IT	2
HR	2

4. SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23 ;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT student_id, name, age, department
FROM students
WHERE department = 'IT' OR department = 'HR' AND age > 21;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	
5	Eve	22	HR

06. SELECT department, COUNT(credits) AS total-credits
 FROM courses
 GROUP BY department
 HAVING total-credits > 5 ;

department	total-credits
IT	11

7. SELECT * FROM courses
 WHERE credits < 4 ;

course-id	course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course-id, course-name, credits
 FROM courses
 ORDER BY credits DESC
 LIMIT 3 ;

course-id	course-name	credit
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. SELECT MAX(grade) AS max-grade, MIN(grade) AS min-grade,
 AVG(grade) AS avg-grade
 FROM enrollments;

max-grade	min-grade	avg-grade
90	78	84.6

10. SELECT course-id, COUNT(enrollment id) AS enrollment-count
 FROM enrollments
 GROUP BY course-id;

course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

11. SELECT SUM(salary) AS total-salary, SUM(bonus) AS total-bonus,
 department
 FROM salaries
 GROUP BY department;

department	total-salary	total-bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12. SELECT department, AVG(salary) AS avg-salary
 FROM salaries
 GROUP BY department
 HAVING avg-salary > 55000 ;

department	avg-salary
IT	61 000
Finance	70 000

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13. SELECT employee-id, name, salary, bonus, salary + bonus AS
 total-compensation
 FROM salaries
 WHERE salary + bonus > 60000 ;

employee-id	name	salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyler	62000	5000	67000

14. SELECT department, SUM(budget) AS total-budget, AVG(budget) AS
 avg-budget
 FROM projects
 GROUP BY department
 HAVING avg-budget > 70000 ;

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	80 000

IS SELECT project-id, project-name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000 AND department
NOT = 'Marketing';

project-id	project-name	department	budget
1	HR App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000